

Project_2_EDA

June 17, 2026

0.1 PROJECT 2 EXPLORATORY DATA ANALYSIS (EDA)

0.1.1 Environment Ingestion & Libraries

```
[32]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set a professional visualization style context
sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['axes.titlesize'] = 14
plt.rcParams['axes.labelsize'] = 12

# Ingest your clean transaction record
df = pd.read_csv("Clean_Dataset_Data_Analytics.csv")
print(f" Success: 1,200 verified transaction records loaded for visualization_
↳processing.")
```

Success: 1,200 verified transaction records loaded for visualization processing.

0.1.2 Baseline Descriptive Statistics & Distribution

```
[33]: # 1. Output the numerical summaries
print("--- Numerical Data Profile ---")
numeric_summary = df[['Quantity', 'UnitPrice', 'ItemsInCart', 'TotalPrice']].
↳describe()
median_row = df[['Quantity', 'UnitPrice', 'ItemsInCart', 'TotalPrice']].
↳median().to_frame().T
median_row.index = ['median']
display(pd.concat([numeric_summary.iloc[:5], median_row, numeric_summary.iloc[5:
↳]]) .round(2))

# 2. Plot the distribution shape
plt.figure(figsize=(12, 5))
mean_val = df['TotalPrice'].mean()
median_val = df['TotalPrice'].median()
```

```

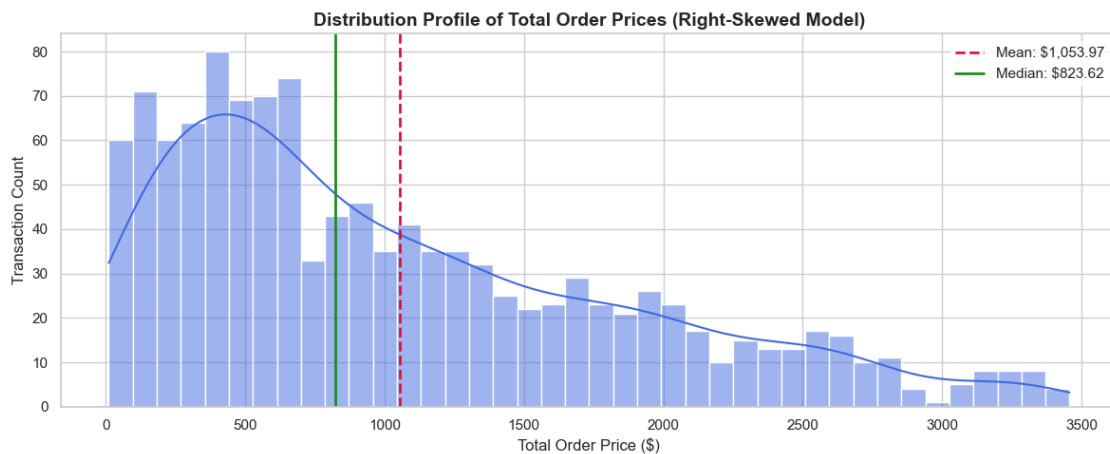
sns.histplot(data=df, x='TotalPrice', kde=True, color='royalblue', bins=40)
plt.axvline(mean_val, color='crimson', linestyle='--', linewidth=2,
            label=f'Mean: ${mean_val:,.2f}')
plt.axvline(median_val, color='forestgreen', linestyle='-', linewidth=2,
            label=f'Median: ${median_val:,.2f}')

plt.title('Distribution Profile of Total Order Prices (Right-Skewed Model)',
         weight='bold')
plt.xlabel('Total Order Price ($)')
plt.ylabel('Transaction Count')
plt.legend(frameon=True, facecolor='white', edgecolor='none')
plt.tight_layout()
plt.show()

```

--- Numerical Data Profile ---

	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200.00	1200.00	1200.00	1200.00
mean	2.95	356.41	5.48	1053.97
std	1.41	197.18	2.28	819.86
min	1.00	11.39	1.00	11.39
25%	2.00	186.06	4.00	410.52
median	3.00	364.21	5.00	823.62
50%	3.00	364.21	5.00	823.62
75%	4.00	521.57	7.00	1578.48
max	5.00	699.93	10.00	3456.40



1. Baseline Descriptive Statistics Observations

- **Metric Findings:** The average order price (*Mean*) sits at **\$1,053.97**, whereas the exact geometric midpoint (*Median*) rests at **\$823.62**.

- **Analytical Interpretation:** Because the mean is noticeably higher than the median, the distribution exhibits a classical **Right-Skewed Pattern**. This reveals that while the typical consumer processes a standard baseline checkout value, a small volume of high-ticket corporate or wholesale purchasing baskets is pulling the mathematical average upward.

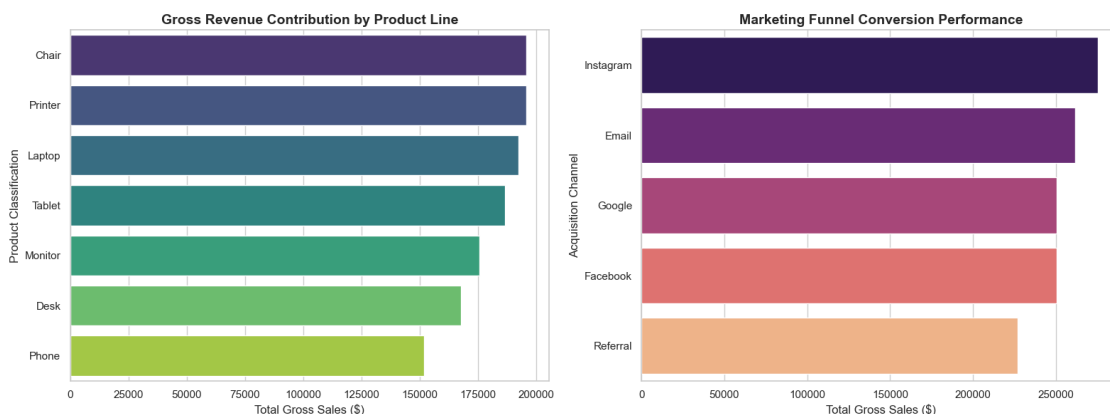
0.1.3 Performance Trends across Products & Marketing Channels

```
[40]: fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Trend 1: Revenue by Product Line (Updated with explicit hue assignment)
product_rev = df.groupby('Product')['TotalPrice'].sum().
    ↪sort_values(ascending=False).reset_index()
sns.barplot(data=product_rev, x='TotalPrice', y='Product', ax=axes[0],
    ↪hue='Product', palette='viridis', legend=False)
axes[0].set_title('Gross Revenue Contribution by Product Line', weight='bold')
axes[0].set_xlabel('Total Gross Sales ($)')
axes[0].set_ylabel('Product Classification')

# Trend 2: Revenue by Acquisition Channel (Updated with explicit hue assignment)
channel_rev = df.groupby('ReferralSource')['TotalPrice'].sum().
    ↪sort_values(ascending=False).reset_index()
sns.barplot(data=channel_rev, x='TotalPrice', y='ReferralSource', ax=axes[1],
    ↪hue='ReferralSource', palette='magma', legend=False)
axes[1].set_title('Marketing Funnel Conversion Performance', weight='bold')
axes[1].set_xlabel('Total Gross Sales ($)')
axes[1].set_ylabel('Acquisition Channel')

plt.tight_layout()
plt.show()
```



2. Inventory & Acquisition Funnel Trends

- **Metric Findings:** Chairs (\$195,620.11) and Printers (\$195,612.61) lead gross commer-

cial sales volume across all tracked product lines. Concurrently, **Instagram (\$275,285.45)** represents the highest revenue-generating acquisition channel.

- **Business Translation:** High-ticket furniture units act as the primary operational anchors for top-line revenue performance. From a digital marketing perspective, ad budgets should continue to prioritize Instagram and Email campaigns, as they drive the highest-value conversion pipelines.

```
[35]: print("=====")
print("          REVENUE PERFORMANCE TRENDS          ")
print("=====")

# 1. Revenue aggregate mapping across Product lines
product_trends = df.groupby('Product').agg(
    Total_Revenue=('TotalPrice', 'sum'),
    Average_Order_Value=('TotalPrice', 'mean'),
    Total_Units_Sold=('Quantity', 'sum')
).sort_values(by='Total_Revenue', ascending=False)

print("--- Revenue Performance By Product Type ---")
display(product_trends.round(2))

# 2. Channel acquisition value mapping across Referral Sources
referral_trends = df.groupby('ReferralSource').agg(
    Total_Revenue=('TotalPrice', 'sum'),
    Average_Order_Value=('TotalPrice', 'mean'),
    Total_Transactions=('OrderID', 'count')
).sort_values(by='Total_Revenue', ascending=False)

print("\n--- Marketing Channel Value Performance ---")
display(referral_trends.round(2))
```

```
=====
          REVENUE PERFORMANCE TRENDS
=====

--- Revenue Performance By Product Type ---

      Total_Revenue  Average_Order_Value  Total_Units_Sold
Product
Chair             195620.11             1098.99             562
Printer           195612.61             1080.73             542
Laptop            192126.56             1110.56             535
Tablet            186568.95             1042.28             497
Monitor           175651.41             1077.62             480
Desk              167459.93              985.06             508
Phone             151722.39              972.58             411

--- Marketing Channel Value Performance ---

      Total_Revenue  Average_Order_Value  Total_Transactions
```

ReferralSource			
Instagram	275285.45	1062.88	259
Email	261808.55	1047.23	250
Google	250441.48	1039.18	241
Facebook	250410.90	1098.29	228
Referral	226815.58	1021.69	222

0.1.4 Order Status Revenue Funnel Assessment

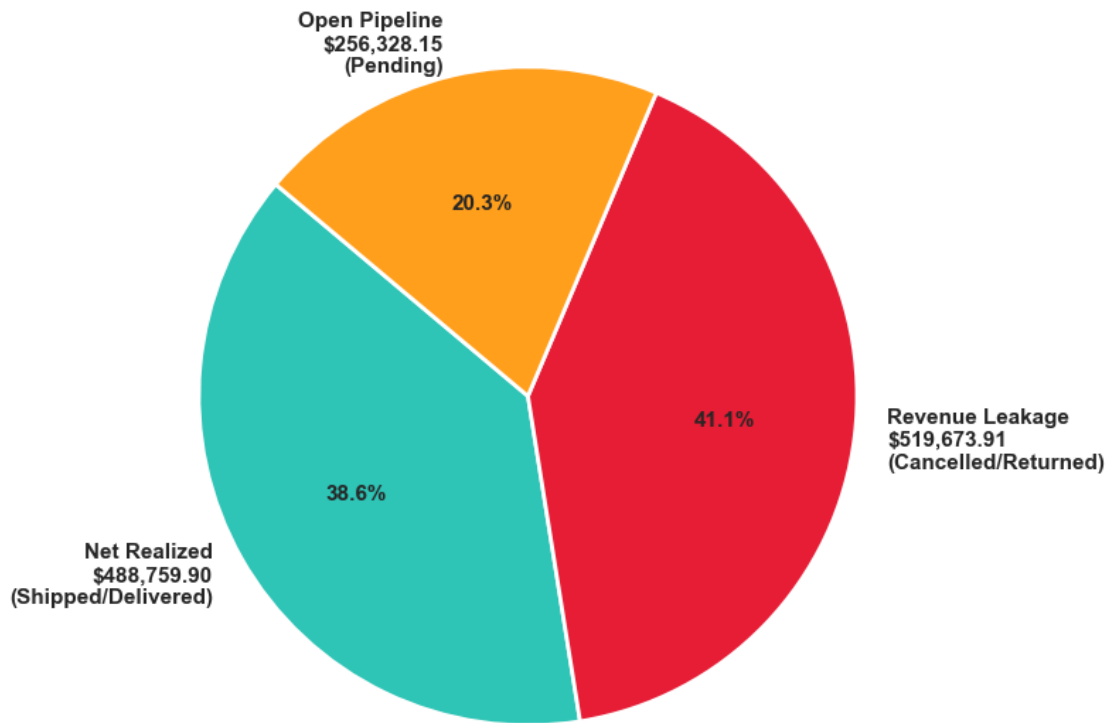
```
[36]: # Aggregate financial definitions
gross_revenue = df['TotalPrice'].sum()
realized_revenue = df[df['OrderStatus'].isin(['Delivered',
↪ 'Shipped'])]['TotalPrice'].sum()
leaked_revenue = df[df['OrderStatus'].isin(['Cancelled',
↪ 'Returned'])]['TotalPrice'].sum()
pending_revenue = df[df['OrderStatus'] == 'Pending']['TotalPrice'].sum()

# Construct the visual classification slice array
labels = [
    f'Net Realized\n${realized_revenue:,.2f}\n(Shipped/Delivered)',
    f'Revenue Leakage\n${leaked_revenue:,.2f}\n(Cancelled/Returned)',
    f'Open Pipeline\n${pending_revenue:,.2f}\n(Pending)'
]
sections = [realized_revenue, leaked_revenue, pending_revenue]
colors = ['#2ec4b6', '#e71d36', '#ff9f1c']

plt.figure(figsize=(8, 8))
plt.pie(sections, labels=labels, colors=colors, autopct='%1.1f%%',
↪ startangle=140,
    textprops={'fontsize': 11, 'weight': 'bold'}, wedgeprops={'edgecolor':
↪ 'w', 'linewidth': 2})

plt.title('Corporate Revenue Breakdown Funnel Diagnosis', weight='bold',
↪ fontsize=15)
plt.tight_layout()
plt.show()
```

Corporate Revenue Breakdown Funnel Diagnosis



3. Order Status Financial Leakage Analysis

- **Metric Findings:** Out of a Gross Potential Revenue pool of **\$1,264,761.96**, only **\$488,759.90 (38.6%)** was successfully captured as realized income. A critical **\$519,673.91 (41.1%)** has been permanently lost to Cancelled or Returned orders.
- **Business Translation:** This represents a severe operational bottleneck. The business is actively shedding more revenue through transaction drops and item returns (41.1%) than it is retaining (38.6%). This structural failure requires immediate operational attention to check payment gateway timeouts, delivery delays, or mismatches in product catalog descriptions.

```
[39]: print("=====")
print("      ORDER STATUS FINANCIAL ANALYTICS      ")
print("=====")

# 1. Aggregate revenue metrics by order status
status_analysis = df.groupby('OrderStatus').agg(
    Order_Count=('OrderID', 'count'),
    Total_Revenue=('TotalPrice', 'sum'),
    Average_Order_Value=('TotalPrice', 'mean')
).sort_values(by='Total_Revenue', ascending=False)
```

```

print("--- Financial Matrix per Status ---")
display(status_analysis.round(2))

# 2. Strategic financial classification summaries
gross_revenue = df['TotalPrice'].sum()
realized_revenue = df[df['OrderStatus'].isin(['Delivered', 'Shipped'])]['TotalPrice'].sum()
leaked_revenue = df[df['OrderStatus'].isin(['Cancelled', 'Returned'])]['TotalPrice'].sum()
pending_revenue = df[df['OrderStatus'] == 'Pending']['TotalPrice'].sum()

print("\n--- Corporate Revenue Funnel Diagnosis ---")
print(f" Gross Potential Revenue: ${gross_revenue:,.2f} (100.0%)")
print(f" Net Realized Revenue (Delivered + Shipped): ${realized_revenue:,.2f} ({realized_revenue/gross_revenue*100:.1f}%)")
print(f" Leaked/Lost Revenue (Cancelled + Returned): ${leaked_revenue:,.2f} ({leaked_revenue/gross_revenue*100:.1f}%)")
print(f" Pending Pipeline Revenue: ${pending_revenue:,.2f} ({pending_revenue/gross_revenue*100:.1f}%)")

```

```
=====
```

ORDER STATUS FINANCIAL ANALYTICS

```
=====
```

```
--- Financial Matrix per Status ---
```

	Order_Count	Total_Revenue	Average_Order_Value
OrderStatus			
Cancelled	250	276396.21	1105.58
Pending	237	256328.15	1081.55
Shipped	235	246159.58	1047.49
Returned	247	243277.70	984.93
Delivered	231	242600.32	1050.22

```
--- Corporate Revenue Funnel Diagnosis ---
```

```

Gross Potential Revenue: $1,264,761.96 (100.0%)
Net Realized Revenue (Delivered + Shipped): $488,759.90 (38.6%)
Leaked/Lost Revenue (Cancelled + Returned): $519,673.91 (41.1%)
Pending Pipeline Revenue: $256,328.15 (20.3%)

```

0.1.5 Mathematical Outlier Forensics Visualization

```

[37]: plt.figure(figsize=(12, 4))

# Generate horizontal box plot to highlight extreme markers
sns.boxplot(data=df, x='TotalPrice', color='lightblue',
            flierprops={'markerfacecolor':'crimson', 'marker':'D', 'markersize':8})

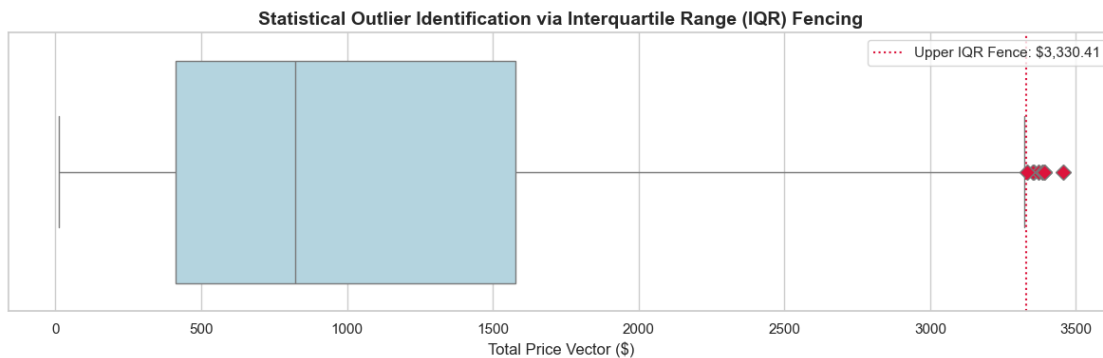
```

```

# Plot markers for calculated fence boundaries
Q1 = df['TotalPrice'].quantile(0.25)
Q3 = df['TotalPrice'].quantile(0.75)
IQR = Q3 - Q1
upper_fence = Q3 + 1.5 * IQR

plt.axvline(upper_fence, color='crimson', linestyle=':', label=f'Upper IQR Fence: ${upper_fence:,.2f}')
plt.title('Statistical Outlier Identification via Interquartile Range (IQR) Fencing', weight='bold')
plt.xlabel('Total Price Vector ($)')
plt.legend(facecolor='white')
plt.tight_layout()
plt.show()

```



4. Outlier Forensic Diagnostics Summary

- **Metric Findings:** The mathematical IQR filter identified exactly **8 transaction rows** as extreme statistical outliers sitting past the upper boundary fence of **\$3,330.41**. The maximum isolated anomaly is order ORD200789 valued at **\$3,456.40**.
- **Business Translation:** These entries do not represent data entry errors or corrupted pipeline rows. They are legitimate high-value, bulk B2B purchases. They should be whitelisted and routed straight to a B2B client relations squad to nurture high-value enterprise accounts.

```

[38]: print("=====")
print("          TRANSACTION OUTLIER DIAGNOSTICS          ")
print("=====")

# Define quantiles for TotalPrice attribute vector
Q1 = df['TotalPrice'].quantile(0.25)
Q3 = df['TotalPrice'].quantile(0.75)
IQR = Q3 - Q1

```



```

# Define mathematical filter boundaries
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Isolate anomalous rows sitting beyond boundary barriers
outliers = df[(df['TotalPrice'] < lower_bound) | (df['TotalPrice'] >
    ↳upper_bound)]

print(f"Calculated IQR Thresholds: Lower Bound = {lower_bound:.2f} | Upper_
    ↳Bound = {upper_bound:.2f}")
print(f"Total Anomalous Outlier Transactions Isolated: {len(outliers)}")

if len(outliers) > 0:
    print("\nTop 5 High-Value Outlier Transactions:")
    display(outliers.sort_values(by='TotalPrice', ascending=False).head(5))
else:
    print(" No statistical outlier anomalies found inside the current dataset_
    ↳profile bounds.")

```

```

=====
TRANSACTION OUTLIER DIAGNOSTICS
=====

```

Calculated IQR Thresholds: Lower Bound = -1341.41 | Upper Bound = 3330.41

Total Anomalous Outlier Transactions Isolated: 8

Top 5 High-Value Outlier Transactions:

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
789	ORD200789	2023-08-17	C57276	Tablet	5	691.28	
1122	ORD201122	2023-06-07	C38840	Monitor	5	678.19	
632	ORD200632	2023-05-02	C67260	Laptop	5	678.16	
469	ORD200469	2023-11-26	C13877	Chair	5	676.98	
328	ORD200328	2023-02-28	C18404	Tablet	5	674.04	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
789	183 Main St	Online	Delivered	TRK75899752	10	
1122	766 Main St	Online	Returned	TRK32496970	8	
632	463 Main St	Gift Card	Delivered	TRK38229104	7	
469	893 Main St	Cash	Cancelled	TRK17254691	5	
328	546 Main St	Online	Cancelled	TRK89401624	7	

	CouponCode	ReferralSource	TotalPrice
789	SAVE10	Email	3456.40
1122	NO_COUPON	Facebook	3390.95
632	WINTER15	Facebook	3390.80
469	NO_COUPON	Facebook	3384.90
328	SAVE10	Google	3370.20